Package 'BPSMKL'

June 29, 2026

Title Bayesian Pathway-Structured Multiple Kernel Kearning (BPS-MKL)

Version 0.0.0.9000

Description BPSMKL is an R package for simultaneously identifying important groups, features within groups, and interactions among features.

License GPL (>= 3)

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2.9000

URL https://github.com/Yuzi896/BPSMKL

BugReports https://github.com/Yuzi896/BPSMKL/issues

LinkingTo Rcpp,
RcppArmadillo

Imports Rcpp

Suggests ggplot2,
patchwork,
pROC

R topics documented:

BPSMKL-package *BPSMKL package*

Description

Bayesian pathway sparse multiple kernel learning.

Author(s)

Maintainer: First Last <first.last@example.com>

See Also

Useful links:

- https://github.com/Yuzi896/BPSMKL
- Report bugs at https://github.com/Yuzi896/BPSMKL/issues

bps_mkl *Run BPS-MKL from individual initial values*

Description

`bps_mkl()` collects the individual initial values into the list expected by the C++ sampler and then runs BPS-MKL algorithm.

Usage

```
bps_mkl(
  iterations,
  burn,
  L,
  As,
  Y,
  X,
  P_eta = 0.2,
  P_gamma = 0.2,
  sigma2 = 0.1,
  lambdas = NULL,
  betas = NULL,
  gammas = NULL,
  Etas = NULL
)
```

Arguments

iterations	Integer. Total number of MCMC iterations to run.
burn	Integer. Number of burn-in iterations used for proposal adaptation.
L	Integer. Number of pathways.
As	List of L numeric adjacency matrices. Each matrix defines the candidate main effects and interactions for one pathway.
Y	Numeric response vector or n x 1 matrix.
X	Numeric n x p design matrix, with observations in rows and predictors in columns.
P_eta	Numeric scalar. eta inclusion probability.
P_gamma	Numeric scalar. gamma inclusion probability.
sigma2	Numeric scalar. initial residual variance.
lambdas	Numeric vector of length 2 * L, with interaction and main-effect bandwidths for each pathway.
betas	Numeric vector of length L containing initial beta values.
gammas	Numeric vector of length L containing initial gamma values.
Etas	List of L numeric eta matrices.

Value

A list with MCMC draws and diagnostics:

gamma_save An iterations by L matrix of gamma draws.

beta_save An iterations by L matrix of beta draws.

sigma_save A numeric vector of sigma2 draws.

eta_whole A list of L matrices storing eta draws for the candidate terms in each pathway.

eta_idx A list of L two-column matrices giving the one-based predictor indices corresponding to the columns of eta_whole.

lambda_save An iterations by 2 * L matrix of lambda draws.

ll_save A numeric vector of log-likelihood values.

time Time used to run the algorithm, returned as a difftime object.

Examples

```
## Not run:
set.seed(1)
L <- 3
n <- 100
p <- 50
sigma2 <- 1
beta <- c(0, 2, 2)
gamma <- ifelse(beta == 0, 0, 1)

pathway <- simulate_pathway(
  p = p,
  n_pathways = L,
  pathway_p = 10,
  n_edges = 15,
```

```
  n_imp_edges = 5,
  n_imp_main = 2,
  main = TRUE,
  seed = 1
)

X <- matrix(runif(n * p), nrow = n, ncol = p)
X <- scale(X)

A <- pathway$A
A_imp <- pathway$A_imp
FX <- matrix(0, nrow = n, ncol = L)

for (l in seq_len(L)) {
  K <- get_K(X, A_imp[[l]], c(1, 1))
  # Sample fx from MVN(0, K).
  FX[, l] <- drop(t(chol(K + 1e-8 * diag(n))) %*% rnorm(n))
}

Y <- drop(FX %*% beta + rnorm(n, sd = sqrt(sigma2)))

result <- bps_mkl(
  iterations = 5000,
  burn = 1000,
  L = L,
  As = A,
  Y = Y,
  X = X,
  sigma2 = sigma2,
  betas = beta,
  gammas = gamma,
  Etas = A_imp
)

## End(Not run)
```

`bps_mkl_app` *Run approximate BPS-MKL from individual initial values*

Description

`bps_mkl_app()` collects the individual initial values into the list expected by the optimized C++ sampler and then runs the approximate Nyström BPS-MKL algorithm.

Usage

```
bps_mkl_app(
  iterations,
  burn,
  L,
  As,
  Y,
  X,
```

```
  m,
  P_eta = 0.2,
  P_gamma = 0.2,
  sigma2 = 0.1,
  lambdas = NULL,
  betas = NULL,
  gammas = NULL,
  Etas = NULL
)
```

Arguments

iterations	Integer. Total number of MCMC iterations to run.
burn	Integer. Number of burn-in iterations used for proposal adaptation.
L	Integer. Number of pathways.
As	List of L numeric adjacency matrices. Each matrix defines the candidate main effects and interactions for one pathway.
Y	Numeric response vector or n x 1 matrix.
X	Numeric n x p design matrix, with observations in rows and predictors in columns.
m	Integer. Number of observations, or rows of X, to sample as Nyström landmarks.
P_eta	Numeric scalar. eta inclusion probability.
P_gamma	Numeric scalar. gamma inclusion probability.
sigma2	Numeric scalar. initial residual variance.
lambdas	Numeric vector of length 2 * L, with interaction and main-effect bandwidths for each pathway.
betas	Numeric vector of length L containing initial beta values.
gammas	Numeric vector of length L containing initial gamma values.
Etas	List of L numeric eta matrices.

Value

A list with MCMC draws and diagnostics:

gamma_save An iterations by L matrix of gamma draws.

beta_save An iterations by L matrix of beta draws.

sigma_save A numeric vector of sigma2 draws.

eta_whole A list of L matrices storing eta draws for the candidate terms in each pathway.

eta_idx A list of L two-column matrices giving the one-based predictor indices corresponding to the columns of eta_whole.

lambda_save An iterations by 2 * L matrix of lambda draws.

ll_save A numeric vector of log-likelihood values.

time Time used to run the algorithm, returned as a difftime object.

Examples

```
## Not run:
# Assume simulation objects are already pre-generated.
# See details in the `bps_mkl()` example.
result <- bps_mkl_app(
  iterations = N,
  burn = burnin,
  L = L,
  As = A,
  Y = Y,
  X = X,
  m = 50,
  sigma2 = sigma2,
  betas = beta,
  gammas = gamma,
  Etas = A_imp
)

## End(Not run)
```

`get_K` *Construct a dense kernel matrix for selected effects*

Description

`get_K()` builds the `N` by `N` kernel matrix used by the model from an input design matrix, an effect-selection matrix, and two bandwidth parameters. The candidate main-effect and interaction index matrix is computed internally from `Eta`.

Usage

```
get_K(X, Eta, lambda, update = FALSE, eta = NULL, i = NULL, j = NULL)
```

Arguments

`X`	Numeric matrix with observations in rows and predictors in columns.
`Eta`	Numeric square matrix indicating which main effects and interactions are active. Diagonal entries represent main effects and off-diagonal entries represent interactions.
`lambda`	Numeric vector of length 2. `lambda[1]` is the bandwidth for interaction terms and `lambda[2]` is the bandwidth for main effects.
`update`	Logical flag. Use `FALSE` to compute the kernel from `Eta` as supplied, or `TRUE` to compute it after temporarily updating the `(i, j)` and `(j, i)` entries of `Eta`.
`eta`	Proposed value to use when `update = TRUE`; must be `0` or `1`. Defaults to `NULL`.
`i`	One-based row index in `Eta` to update when `update = TRUE`. Defaults to `NULL`.
`j`	One-based column index in `Eta` to update when `update = TRUE`. Defaults to `NULL`.

Details

Off-diagonal active entries in `Eta` are treated as interactions and use `lambda[1]`; diagonal active entries are treated as main effects and use `lambda[2]`. When `update = TRUE`, the C++ helper works with a temporary copy of `Eta` where `Eta[i, j]` and `Eta[j, i]` are set to `eta`. The candidate index matrix is always computed from the original `Eta`.

Value

An `N` by `N` numeric kernel matrix, where `N = nrow(X)`.

`res_summary` *Summarize BPS-MKL MCMC results*

Description

`res_summary()` summarizes posterior pathway and eta inclusion probabilities from a fitted BPS-MKL result object. It always returns an `idx_res` data frame with pathway labels, predictor index pairs, and estimated eta probabilities. When simulated truth is supplied through `pathway`, it also returns pathway-level AUC values. When `Figure = TRUE` and `ggplot2` is available, it also returns diagnostic heatmaps.

Usage

```
res_summary(
  res,
  pathway = NULL,
  gammas = NULL,
  p = NULL,
  burnin = 0,
  Figure = FALSE
)
```

Arguments

`res`	Result list returned by `bps_mkl()` or `bps_mkl_app()`.
`pathway`	Optional simulated pathway object, such as the output from `simulate_pathway()`. When supplied, `pathway$A_imp` is used as truth for AUC calculation and optional figures.
`gammas`	Optional numeric vector of true pathway inclusion indicators. When supplied with `pathway` and `Figure = TRUE`, truth heatmaps multiply `A_imp` by `gammas`.
`p`	Optional integer number of predictors. If `NULL`, this is inferred from `dim(res$eta_whole[[1]])[1]`.
`burnin`	Integer number of initial MCMC draws to discard. Defaults to `0`.
`Figure`	Logical. If `TRUE`, return heatmap figures when `ggplot2` is available. Defaults to `FALSE`.

Value

A list containing:

`idx_res` A data frame with `pathway_id`, `idx1`, `idx2`, `estimated`, and `P_pathway`. If `pathway` is supplied, it also contains `truth`.

`auc_sim` A data frame with pathway-level AUC values. Returned only when `pathway` is supplied.

`Figure` A list of pathway heatmaps. Returned only when `Figure = TRUE` and `ggplot2` is available. If `pathway` is supplied, truth heatmaps are included.

`simulate_pathway` *Simulate pathway adjacency matrices*

Description

`simulate_pathway()` creates non-overlapping simulated pathways over `p` features. For each pathway, it samples `pathway_p` features, creates `n_edges` candidate interaction edges among those features, and marks `n_imp_edges` of those candidate edges as important. Optionally, it can also add important main effects on the diagonal of each important-effect matrix.

Usage

```
simulate_pathway(
  p,
  n_pathways,
  pathway_p,
  n_edges,
  n_imp_edges,
  n_imp_main = 0,
  main = FALSE,
  seed = NULL
)
```

Arguments

`p`	Integer. Total number of features.
`n_pathways`	Integer. Number of pathways to simulate.
`pathway_p`	Integer. Number of features assigned to each pathway.
`n_edges`	Integer. Number of candidate interaction edges to include in each pathway adjacency matrix.
`n_imp_edges`	Integer. Number of candidate interaction edges to mark as important in each pathway.
`n_imp_main`	Integer. Number of pathway features to mark as important main effects when `main = TRUE`. Defaults to `0`.
`main`	Logical. If `TRUE`, add important main effects on the diagonal of each important-effect matrix. Defaults to `FALSE`.
`seed`	Optional integer random seed used before simulation. Defaults to `NULL`.

Value

A list with two elements:

`A` A list of `n_pathways` `p x p` adjacency matrices containing the candidate interaction edges for each pathway.

`A_imp` A list of `n_pathways` `p x p` matrices containing the important interaction edges and, when requested, important main effects.

Index